

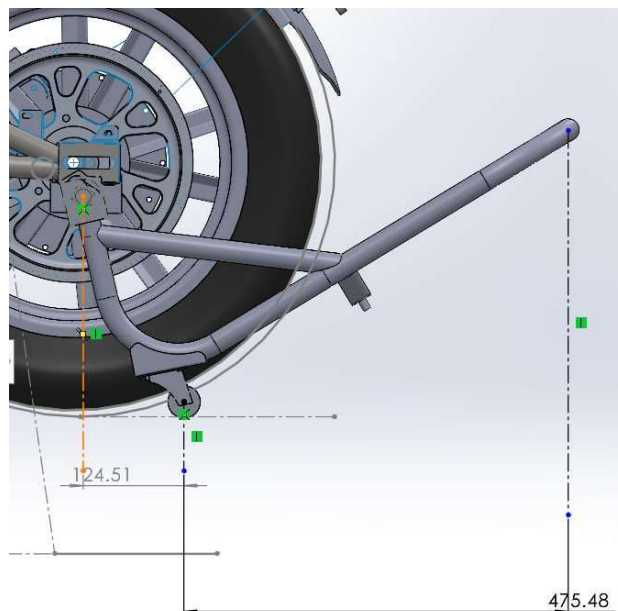
Design and Development of Rear Paddock Stand for the VEGA Street Bike

1. Project Overview

This project was carried out during the industrial training period at Vega Projects, focusing on the design and development of a rear paddock stand for the Vega Street Bike. A paddock stand is an essential workshop and racing tool used to elevate the rear wheel of a motorcycle off the ground, enabling safe and convenient access for maintenance, servicing, chain adjustment, tire inspection, and wheel removal.

2. Design Phase

The design phase began with a study of existing paddock stand designs and an analysis of the load requirements based on the motorcycle's weight. Concept sketches were developed and evaluated against criteria such as material efficiency, manufacturability, and user safety. The selected concept was then modeled in SolidWorks as a detailed 3D CAD assembly, incorporating all structural members, pivot joints, and lifting arms.



3. Simulation and Structural Analysis

Structural analysis of the rear paddock stand was carried out using SolidWorks Simulation, a Finite Element Analysis (FEA) tool integrated within the SolidWorks CAD environment. The purpose of the simulation was to verify that the stand could safely support the rear weight of the Vega Street Bike without experiencing failure, excessive deformation, or unsafe stress concentrations under realistic loading conditions.

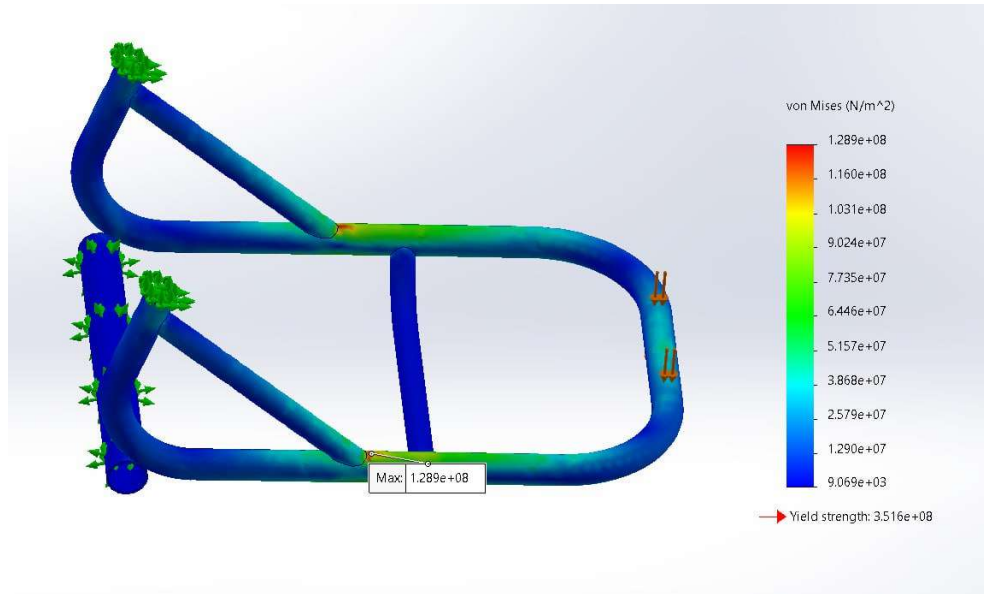


Figure 1 - Von Mises Stress Analysis of the Rear Paddock Stand Frame Under Applied Hand Force

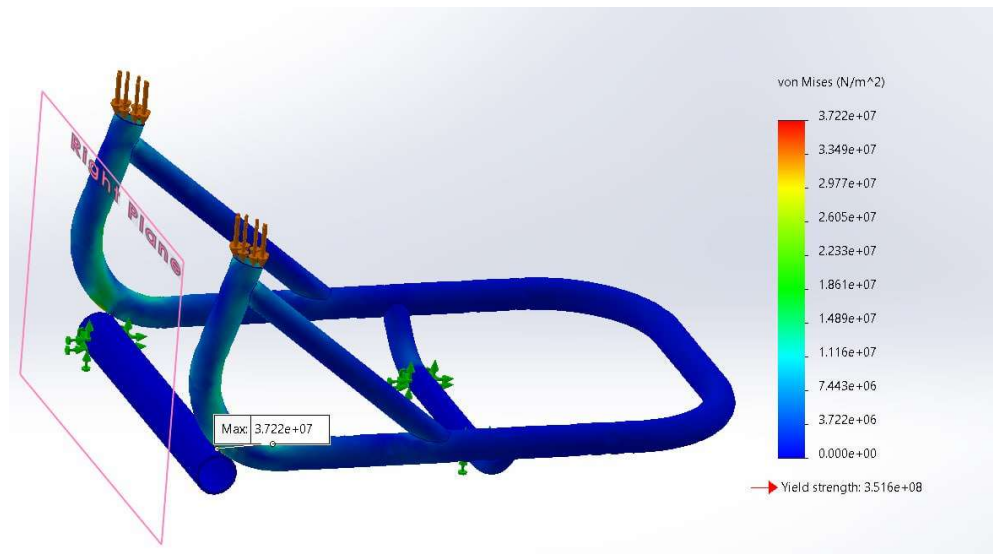


Figure 2 - Von Mises Stress Analysis of the Rear Paddock Stand Assembly Under Motorcycle Rear Weight Loading

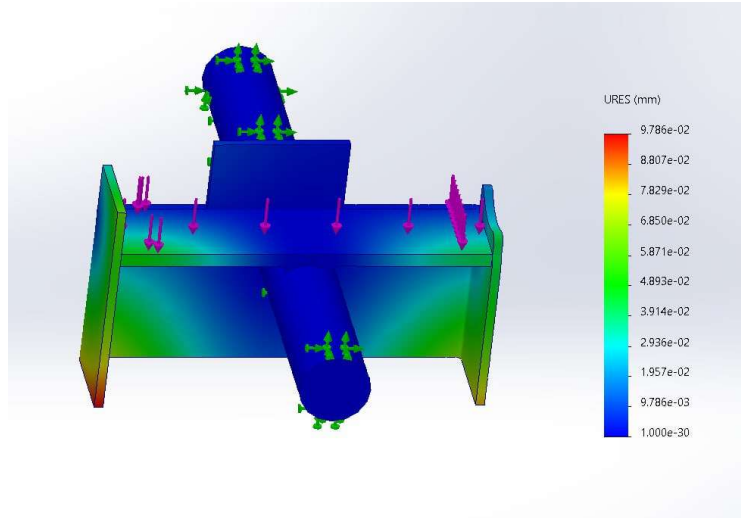


Figure 3 - Resultant Displacement (URES) Analysis of the Paddock Stand Connecting Mount

4. Welding Jig Design for Component Alignment

Prior to fabrication, a welding jig was designed to ensure accurate and consistent positioning of the paddock stand components during the welding process. The jig was developed to hold all structural members in their correct orientations, preventing misalignment and angular distortion caused by welding heat. This ensured dimensional accuracy of the final assembly and repeatability if multiple units were to be produced.

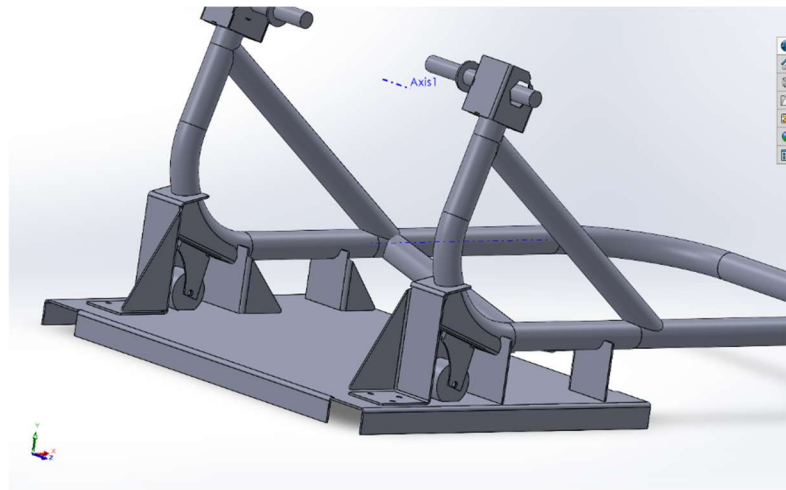


Figure 4 -Welding Jig for fixing the assembly

5. Manufacturing Phase

Following design validation, the paddock stand was fabricated using mild steel hollow sections cut and prepared according to the SolidWorks CAD drawings. Components were positioned using the welding jig to ensure accurate alignment, then joined using MIG welding. After welding, surfaces were ground smooth and the stand was finished with primer and paint for corrosion protection and a clean appearance.



Figure 5 - Lazer cut parts



Figure 6 - Bent tubes



Figure 7 - Paddock Stand Assembly on the Welding Jig

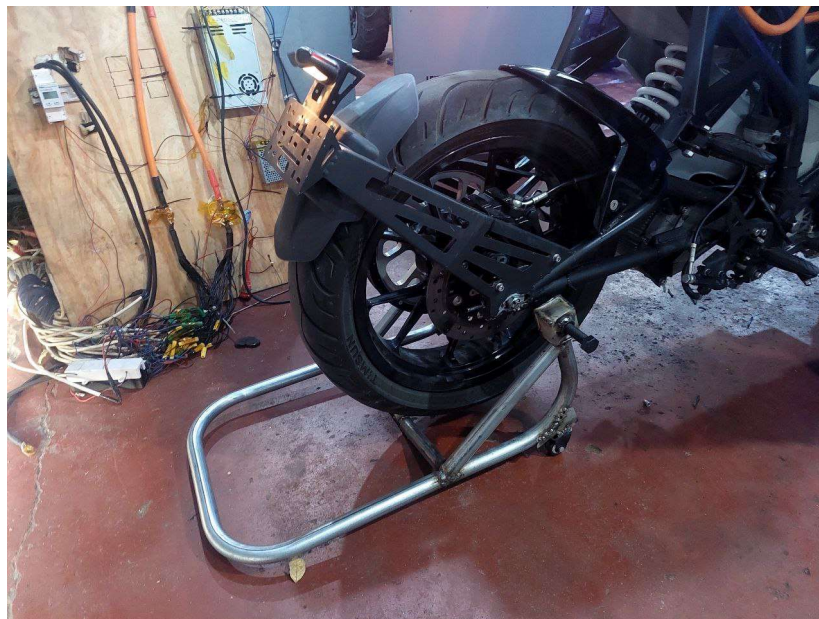


Figure 8 - Fabricated Rear Paddock Stand Supporting the Vega Street Bike During Final Testing